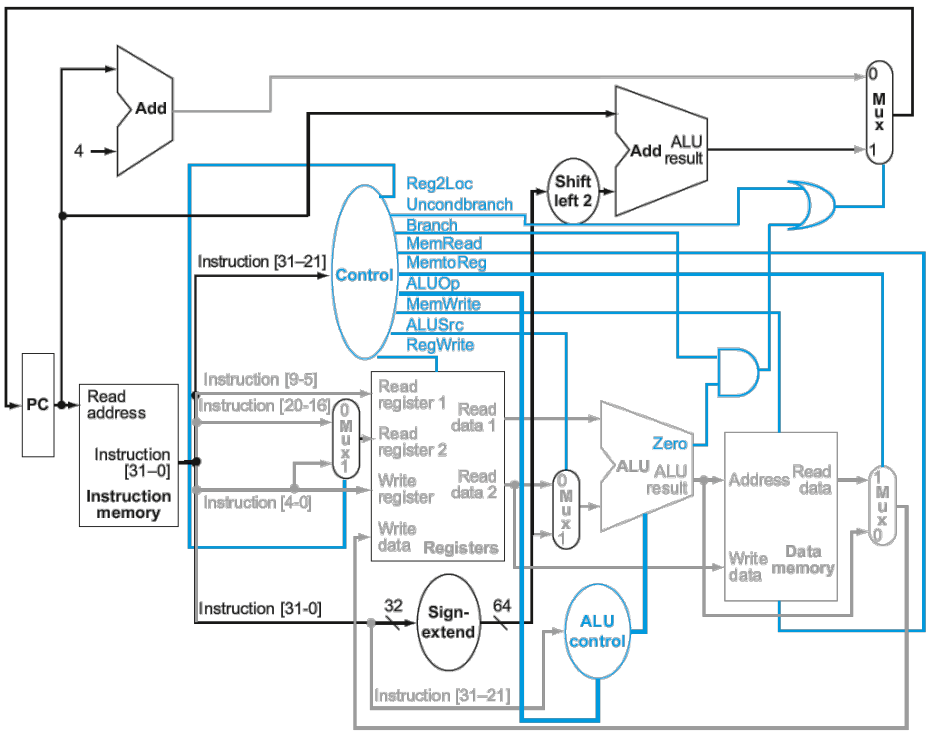
Keys.

CpE 3110/CS 3803 Homework Assignment #4 Maximum – 100pts

You must always show or explain your work in a neat and orderly format. You are encouraged to discuss ideas with other students and consult references but your work must be your own.

Be sure to show your work/derivation for full credit (or maximum possible partial credit, if your final answer is not right). Also, do your best to maximize your HW score, since 13% of your semester letter grade will be decided by your HW performance.

For the single-cycle LEGv8 processor design given below, answer the following questions:



1. List all combinational logic components used in this design. [10pts]

MUX, Sign-extend, shift left2, ALU, control, ALUcontrol, Add, AND, OR

2. List all sequential logic components used in this design. [10pts]

PC, Instruction memory, Register file, Data memory

3. Among the following instructions implemented in this design, which ones may use the 64b output from the sign-extend unit? Clearly justify your answer. [10pts]

Memory reference: LDUR, STUR → For byte offset.
Arithmetic/logical: ~~ADD, SUB~~, AND and ORR
Control transfer: CBZ and B → For instruction offset.

4. Find appropriate values for all control signals (Reg2Loc, Uncondbranch, Branch, MemRead, MemtoReg, 2-bit ALUOp, MemWrite, ALUSrc, RegWrite and 4-bit ALU control) to execute each of the following instructions. Note MemRead/MemWrite should be 00, 10 and 01 for hold/read/write operations, respectively. Also, make sure to use don't cares when possible: [50pts]

```
Loop: ADD   X10, XZR, SP
      ADD   X9, X10, X11
      LDUR  X9, [X10, #-16]
      CBZ   X9, Loop
      STUR  X9, [X10, #-8]
      B     Loop
```

X means don't care.

Inst	Reg2Loc	Uncond branch	Branch	Mem Read	Mem to Reg	ALUOp	Mem write	ALU src	Reg write	ALU operation
ADD	0	0	0	0	0	10	0	0	1	0010
ADD	0	0	0	0	0	10	0	0	1	0010
LDUR	X	0	0	1	1	00	0	1	1	0010
CBZ	1	0	1	0	X	01	0	0	0	0111
STUR	~~X~~ 1	0	0	0	X	00	1	1	0	0010
B	X	1	X	0	X	XX	0	X	0	XXXX

5. For the instruction sequence given in the previous question, calculate the 64b output from the shift left 2 unit for CBZ instruction and B instruction. Clearly show your work. [20pts]

Instruction offset for CBZ is $-3_{10} = \underbrace{1111 \cdots 1101_2}_{64 \text{ bits.}}$

shift left 2 to convert it to byte offset

$\Rightarrow \underbrace{1111 \cdots 1111\ 0100_2}_{64 \text{ bits.}}$

For B inst, inst offset $= -5_{10} = 1111 \cdots 1011_2 \xrightarrow{\text{shift left 2}}$

$1111 \cdots 1110\ 1100_2$.